

Tubulointerstitial Diseases

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KEY POINTS

- Tubulointerstitial diseases are a relatively common cause of acute kidney injury (10%–20% of cases) and are likely to be a significant unrecognized cause of chronic kidney disease (CKD).
- The majority of acute tubulointerstitial nephritis (ATIN) cases are caused by drugs (70%), followed by immunologic factors (20%) and infectious etiologies (4%).
- Antibiotics, proton pump inhibitors, and nonsteroidal antiinflammatory drugs are the most frequent culprits of drug-induced ATIN (DI-ATIN). However, cancer therapies, particularly immune checkpoint inhibitors, are increasingly recognized as potential causes.
- Patients with ATIN present with minimal symptoms and an acute or a subacute rise in serum creatinine. The classic triad of fever, rash, and eosinophilia is uncommon and seen predominantly in drug-induced ATIN due to β -lactams and sulfonamides. Urine sediment findings, urine eosinophils, and imaging tests such as gallium scanning have limited sensitivity and specificity, so a kidney biopsy is generally required to make a definitive diagnosis.
- Early use of steroids is generally recommended for treatment of drug-induced ATIN if there is no improvement after drug discontinuation, but this is based on limited evidence from small, observational studies. There is no standard steroid regimen. Failure to diagnose and treat ATIN early may result in increased kidney fibrosis and incomplete recovery of kidney function.
- Chronic tubulointerstitial nephritis (CTIN) is commonly caused by genetic and toxic factors. Genetic diseases such as autosomal dominant tubulointerstitial kidney diseases are caused by mutations in different genes with *UMOD* and *MUC1* being the most frequent. Other genetic diseases include nephronophthisis, which may contribute to cases of undiagnosed end-stage kidney disease in adulthood, and cystinosis, which is the most common cause of renal Fanconi syndrome in children.
- Toxic causes of CTIN include aristolochic acid, heavy metals, radiation, combined analgesics, and metabolic causes such as hypokalemia, hyperuricemia, and hyperoxaluria.
- CKD of unknown etiology has been reported as a cause of CTIN. It affects young-to-middle-aged men working in agricultural fields, particularly in Sri Lanka and Mesoamerica. Heat stress, exposure to pesticides, consumption of water contaminated with agrochemicals and heavy metals, and genetic predisposition are potential contributors.

EPIDEMIOLOGY

Tubulointerstitial nephritis (TIN) is a condition characterized by inflammation in the kidney interstitium and tubules, resulting in a decline in kidney function. Acute

or immune-mediated tubulointerstitial nephritis (ATIN) refers to the acute form of the disease, which presents with a rapid decline in kidney function, while chronic tubulointerstitial nephritis (CTIN) has a more prolonged course with increased fibrosis observed through histology. The definitive diagnosis of ATIN or CTIN relies on histologic

tissue examination because no reliable or commercially available noninvasive markers exist. Estimating the exact incidence of ATIN and CTIN is challenging due to variations in prevalence and etiologies worldwide, as well as differences in kidney biopsy practices. However, global prevalence studies among patients with acute or chronic kidney disease (CKD) indicate that ATIN affects approximately 3% to 6% of patients undergoing a kidney biopsy.^{1,2}

ATIN accounts for 13% to 19% of acute kidney injury (AKI) cases and is more commonly observed in elderly individuals.^{2,3} Certain populations, such as HIV-positive patients, display a higher incidence of TIN, ranging from 11% in the United States to 13% in French series. On the other hand, the true epidemiology of chronic or nonimmune-mediated tubulointerstitial nephritis remains unknown and is less documented compared with ATIN. Genetic and toxic factors are increasingly recognized causes of CTIN.

MECHANISMS OF TUBULOINTERSTITIAL INJURY

Tubulointerstitial injury can occur through various mechanisms involving innate and adaptive immune responses resulting in inflammation and fibrosis. The tubulointerstitium consists of various intrinsic cells, structural components, and infiltrating cells.^{4,5} It encompasses the intertubular, extraglomerular, and extravascular spaces of the kidney, as well as the renal tubules, blood vessels, and different cell types. These include fibroblasts, macrophages, lymphocytes, lymphatic endothelial cells, and dendritic cells. Interstitial fibroblasts, derived from mesenchymal origin, are the most abundant cell type within the interstitium.⁶ They express proteins that regulate the production and degradation of the extracellular matrix, influencing the amount of interstitial fibrosis.⁷ Causes of tubulointerstitial diseases can be broadly divided into immune- and non-immune-mediated disorders (Table 37.1).

IMMUNE-MEDIATED TUBULOINTERSTITIAL DISEASE

The immune response involves antigen recognition, presentation, immune regulation, and the effector phase characterized by infiltration of inflammatory cells (Fig. 37.1). The immune response can be cell mediated, such as in drug-mediated tubulointerstitial nephritis, or an autoimmune phenomenon, such as antibrush border-mediated tubulointerstitial nephritis.

When the immune system targets the renal interstitial compartment, it recognizes antigens from internal and external sources. Endogenous antigens can include the tubular basement membrane (TBM) and other components of the extracellular matrix, while exogenous antigens are processed and presented by native renal cells or minor populations of dendritic cells within the interstitium.⁵ The proximal TBM antigen is the major target of anti-TBM antibody-mediated interstitial nephritis in humans.⁵

T cells recognize foreign antigens when presented by antigen-presenting cells (APCs) through the binding of MHC antigen molecules. CD4⁺ T cells recognize antigens presented with class II MHC molecules resulting in cell proliferation and cytokine expression. On the other hand,

Table 37.1 Broad Causes of Tubulointerstitial Nephritis

Broad Causes of Tubulointerstitial Nephritis	
Immune Mediated	Nonimmune Mediated
Cell Mediated	Genetics
Drugs	Autosomal dominant tubulointerstitial kidney diseases (ADTKD)
Infection	Nephronophthisis (NPHP)
Kidney transplant rejection	Cystinosis
	Karyomegalic Interstitial Nephritis
Autoimmune	Metabolic
Sjogren	Hypokalemia
IgG4-related disease	Crystals (phosphorus, Ca oxalate, uric acid)
Tubulointerstitial nephritis with uveitis syndrome	Toxic
Tubulointerstitial nephritis with IgM-positive plasma cells (IgMPCTIN)	Heavy metals
Antibrush border antibodies	Lithium
	Cyclosporin
	Radiation
	Analgesics
	Endemic nephropathy (Balkan, CKDu)
	Others
	Vesicoureteral reflux
	Lymphoproliferative disorders, plasma cell dyscrasias
Examples of Systemic Diseases Associated with Secondary Tubulointerstitial Nephritis	
Systemic lupus erythematosus	Diabetes mellitus
Vasculitis	Cystic kidney diseases
Sarcoidosis	Acute kidney injury

CD8⁺ T cells recognize antigens synthesized by APCs in conjunction with class I MHC molecules and cause target cell death.⁵ Proximal tubules have antigen-presenting capability and secrete growth factors and cytokines that determine the interaction between cells and the extracellular matrix.^{5,7}

The interstitial infiltrate found in most human CKDs consists of a number of different effector cells, with CD4⁺ T cells being prominent.⁵ Circulating monocytes that migrate into the kidneys transform into renal interstitial macrophages and can differentiate into two phenotypes: M1 or inflammatory and M2 or reparative. B and T lymphocytes are commonly found alongside macrophages in the interstitium. Because the composition of the interstitial infiltrate is similar regardless of the cause, it is possible that immune responses to injured interstitial cell neoantigens are a final common pathway for interstitial injury. The histologic appearance of interstitial disease can vary, potentially reflecting different stages of immune-mediated damage or the activity of distinct populations of activated T cells.^{5,7}

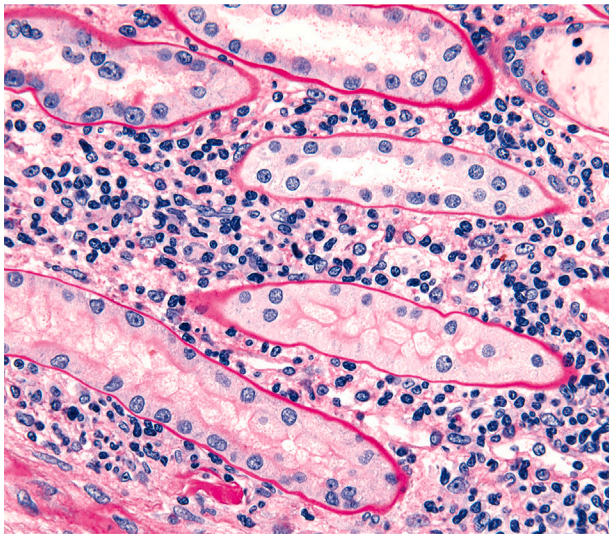


Fig. 37.1 Renal histology observed in acute tubulointerstitial nephritis highlighting the inflammatory infiltrate. Periodic acid-Schiff stain. (From Perazella MA. Clinical approach to diagnosing acute and chronic tubulointerstitial disease. *Adv Chronic Kidney Dis.* 2017;24:57–63.)

PATHOGENESIS OF DRUG-INDUCED ACUTE TUBULOINTERSTITIAL NEPHRITIS

Drug-induced acute tubulointerstitial nephritis (DI-ATIN) is primarily considered a T-cell-mediated delayed hypersensitivity reaction (type IV) occurring 7 to 10 days after exposure.⁵ The kidney susceptibility to hypersensitivity is influenced by factors such as high blood delivery and local drug metabolism. Drugs can induce antigens through various mechanisms. They can bind to circulating or intrinsic tubulointerstitial proteins in a process called haptization (e.g., β -lactam antibiotics bind to lysine residues on circulating proteins, particularly albumin, to form an immunogenic complex); act as prohaptens such as sulfamethoxazole, which is modified by the CYP2C9 isoenzyme in the liver before spontaneously converting to the toxic nitroso-sulfamethoxazole, which then binds to cysteine residues on intracellular and extracellular proteins to form a hapten; or resemble native kidney antigens (molecular mimicry), leading to immune reactions.⁵

Even though DI-ATIN is usually a type IV hypersensitivity reaction, it can cause injury through other pathways, too. Antitubular basement membrane (anti-TBM) TBM antibody deposition can occur with medications such as methicillin, rifampin, allopurinol, and phenytoin, which usually stain positive for IgG and C3 on immunofluorescence in a granular pattern. Nonsteroidal antiinflammatory drugs (NSAIDs) are implicated in a non-type IV drug-hypersensitivity reaction associated with lack of tissue eosinophil infiltrate on kidney biopsy. In addition, drugs and drug-hapten complexes can induce inflammation by directly forming immune deposits within the interstitium or precipitating in the form of circulating complexes. Examples of drugs that can cause this immune response include certain penicillins, cephalosporins, and phenytoin.⁵

The kidney self-regulates the immune response through activation of suppressor T cells and downregulation of MHC class II expression. However, medications like checkpoint

inhibitors used in cancer treatment can disrupt immune regulation. After antigens are presented, lymphocytes, neutrophils, macrophages, mast cells, and eosinophils infiltrate the interstitial space, with T lymphocytes constituting the majority of the infiltrative cells.⁵

NON-IMMUNE-MEDIATED TUBULOINTERSTITIAL DISEASE

In addition to immune-mediated tubulointerstitial injury, a variety of non-immune-mediated processes promote damage to this compartment.

In glomerular diseases, impaired glomerular permselectivity, intraglomerular hypertension, glomerular hypoperfusion, immunologic disturbances, and entry of inflammatory mediators to the interstitium through the juxtaglomerular apparatus lead to worsening inflammation and glomerular ischemia.⁷ The loss of nephrons triggers metabolic adaptations in surviving nephrons, leading to tubulointerstitial injury through the renin-angiotensin system. Misdirected filtration and cell proliferation in the Bowman space are other mechanisms that can cause inflammation and damage in the tubulointerstitial space.⁸

Proteinuria is not only an indicator of glomerular damage but also an independent risk factor for kidney disease progression. Proteinaceous casts and glomerular tuft adhesions can obstruct tubular lumens and cause tubulointerstitial injury.^{7,9} Additionally, filtered proteins have direct effects on tubular cells, particularly in the proximal segments leading to activation of proinflammatory and profibrogenic processes.⁹

Urinary proteins such as ultrafiltered growth factors and cytokines present in the nephrotic tubular fluid stimulate the expression of matrix proteins, leading to interstitial fibrosis.⁹ Also, complement factors originating from the bloodstream or renal tubules, in particular the C5b-9 complex, cause significant damage in the renal tubular epithelial cells because they lack certain membrane-bound complement regulatory proteins.¹⁰ Accumulation of complement within proximal tubular cells is followed by local recruitment of infiltrating mononuclear cells, driving progressive tubulointerstitial damage.

Fibrosis is the final common pathway leading to CKD, irrespective of the initial trigger. It involves the loss of renal tubules and accumulation of fibroblasts and matrix proteins like collagen (types I–V and VII), fibronectin, and laminin.¹¹ Fibroblasts are the principal effector cells responsible for mediating tubulointerstitial fibrosis through a process of epithelial-mesenchymal transition (EMT), which takes place after repeat injury.¹² During EMT, epithelial and endothelial cells undergo a transition to become fibroblasts in response to various growth factors and cytokines, with TGF- β and epidermal growth factors playing significant roles.¹² Importantly, mature epithelia are in a dynamic but not terminal state of differentiation; therefore chronic inflammation into the interstitium destabilizes epithelial tissues and favors fibrogenesis.¹³

Finally, chronic ischemia driven by chronic vasoconstriction mediated by angiotensin II and reduced nitric oxide production stimulate EMT.¹⁴ Rarefaction, the loss of peritubular capillaries, further worsens the oxygen supply to tubular cells and leads to tubular atrophy and

loss.¹⁵ The remaining tubules may experience functional hypermetabolism resulting in increased oxygen consumption and the progression of tubulointerstitial fibrosis.

ACUTE TUBULOINTERSTITIAL NEPHRITIS

The majority of ATIN cases are caused by drugs (70%), followed by immunologic factors (20%) and infectious etiologies (4%)¹⁶ (Table 37.2).

ATIN is often incidentally detected through elevated serum creatinine levels and abnormalities observed in urinalysis, such as sterile pyuria, tubular proteinuria, microscopic hematuria, and the presence of white blood cell (WBC) and/or red blood cell (RBC) casts. The most common urinary finding is low-grade proteinuria, typically ranging from 0.3 to 1 g/day, while the classic triad of fever, morbilliform rash affecting the trunk, and eosinophilia is observed in less than 10% of patients.¹⁷ It is important to note that the presence of urine eosinophils, evaluated using both 1% and 5% cutoff thresholds, has been reported across various AKI diagnoses and is not recommended for diagnosing ATIN.¹⁶ We will review specific causes of ATIN and their distinct features.

DRUGS

Medications are the leading cause of ATIN, accounting for more than 70% of cases in developed countries and approximately 50% in developing countries.^{16,18,19} Drug-induced ATIN (DI-ATIN) is a dose-independent type IV hypersensitivity reaction that affects the interstitial space after exposure to different agents. The list of medications associated with ATIN is constantly expanding, but common culprits include antimicrobial agents, proton pump inhibitors (PPIs), NSAIDs, antiepileptics, and anticancer drugs. Antibiotics, PPIs, and NSAIDs are the most frequent culprits, with varying frequencies depending on different reports. Older adults are at a higher risk due to polypharmacy and preexisting kidney disease. Discontinuing the offending agent is crucial but can be challenging, particularly because in up to one third of the cases, the culprit drug is not identified.¹⁶

Studies have shown that antibiotics are responsible for approximately 49% of DI-ATIN cases, while PPIs and NSAIDs contribute to 14% and 11% of cases, respectively.¹⁶ However, different studies conducted in New Zealand and the United Kingdom identified PPIs as the primary cause of ATIN in a significant percentage of biopsy-proven cases, ranging from 57% to 80%.^{20,21} Immune checkpoint inhibitors (ICPIs), powerful monoclonal antibodies used in cancer treatment, have also been associated with AKI. Interstitial nephritis was the dominant lesion in 93% of 138 biopsied patients with immune checkpoint inhibitor–associated AKI.²²

ANTIMICROBIALS

Certain antibiotics such as β -lactams (penicillins and cephalosporins), sulfonamides, and fluoroquinolones have been associated with ATIN. β -lactam antibiotics can lead to ATIN within a few days to a few weeks after antibiotic exposure. A hypersensitivity syndrome characterized by fever, morbilliform rash affecting the trunk, or eosinophilia

Table 37.2 Specific Causes of Acute Tubulointerstitial Nephritis

Class	Specific Causative Agents
Drugs 70%	
Antimicrobials	β -Lactams (penicillin and derivatives, cephalosporins), quinolones, ethambutol, isoniazid, macrolides, rifampin, sulfonamides, tetracycline, vancomycin, antiviral agents (acyclovir, foscarnet, indinavir, and atazanavir)
NSAIDs and COX-2 inhibitors	Almost all agents
Gastrointestinal drugs	Proton pump inhibitors, H_2 antagonists, mesalamine, sulfasalazine
Diuretics	Furosemide, thiazides, triamterene
Anticancer agents	BRAF inhibitors, bortezomib, ifosfamide, tyrosine kinase inhibitors, pemetrexed, immune checkpoint inhibitors, lenalidomide, platinum salts, etc.
Miscellaneous	Allopurinol, amlodipine, azathioprine, captopril, carbamazepine, clofibrate, cocaine, creatine, diltiazem, phentermine, phenytoin, pranlukast, propylthiouracil, quinine, phenindione, synthetic cannabinoids
Systemic Diseases 20%	
Immune	SLE tubulointerstitial nephritis, sarcoidosis, Sjögren syndrome, IgG4-related kidney disease, TINU, ANCA-related diseases, spontaneous tubulointerstitial nephritis
Infectious Agents 4%	
Bacteria	<i>Corynebacterium diphtheriae</i> , <i>Escherichia coli</i> , <i>Legionella</i> , <i>Staphylococcus</i> , <i>Streptococcus</i> , <i>Yersinia</i> , <i>Brucella</i> , <i>Campylobacter</i> , <i>Legionella</i>
Viruses	Cytomegalovirus, Epstein-Barr virus, hantaviruses, hepatitis C, herpes simplex virus, HIV, polyoma virus, adenovirus
Others	<i>Leptospira</i> , <i>Mycobacterium</i> , <i>Mycoplasma</i> , <i>Chlamydia</i> , rickettsia, syphilis, toxoplasmosis, fungi
Others	
Transplant	Acute T-cell–mediated rejection (ATCR)
Neoplastic	Lymphoproliferative disorders, plasma cell dyscrasias

is commonly observed with β -lactam antibiotics.²³ About 75% of affected patients experience urinary abnormalities such as proteinuria, pyuria, or hematuria.¹⁶

Sulfonamides tend to affect transplant recipients, HIV patients, and those with reduced kidney function more frequently due to their higher usage in these populations. Hypersensitivity syndromes are also common in this group.²³

Fluoroquinolones, especially ciprofloxacin, are commonly reported as a cause of DI-ATIN, likely due to their widespread use.¹⁶ Rifampin can induce dose-dependent ATIN accompanied by antirifampin antibodies, hemolytic anemia, thrombocytopenia, and hepatitis.²⁴ Additionally, various antiviral agents have been associated with ATIN, especially in the HIV population.²⁵

PROTON PUMP INHIBITORS

The use of proton pump inhibitors (PPIs) for acid-related gastrointestinal disorders has been steadily increasing, and their association with ATIN has gained recognition.^{26,27} According to a 10-year retrospective study in Australia, 64% of biopsy-confirmed cases of ATIN were linked to the use of PPIs. This kidney injury usually occurred 11 weeks after starting PPI treatment.²⁸ In New Zealand, a population-based study revealed that PPI users had a fivefold higher risk of developing ATIN compared with nonusers, with the risk being particularly elevated in patients older than the age of 60.²⁷ A similar study conducted in Ontario residents older than 66 years of age found a threefold increased hazard ratio for AKI and ATIN, requiring hospitalization within 120 days of initiating PPI therapy.²⁹

Diagnosing PPI-induced ATIN can be challenging because it often lacks systemic allergic symptoms. In fact, only a quarter of cases were suspected before kidney biopsy in a U.S. case series.³⁰ PPI-induced ATIN generally has a favorable prognosis with 28.5% of cases showing complete recovery, 43% partial recovery, and 28.5% no recovery at 6 months' follow-up.³⁰ End-stage kidney failure (ESKF) has been reported in 11% to 12% of the cases.^{21,28}

The use of PPIs has been linked to CKD, and it has been suggested that untreated episodes of AKI or ATIN might be the underlying cause.³¹

NONSTEROIDAL ANTIINFLAMMATORY AGENTS

NSAIDs including selective and nonselective cyclooxygenase (COX) inhibitors are commonly used for pain, fever, and inflammation. While their main drawback is gastrointestinal side effects, they can also cause various nephrotoxic effects, with hemodynamic AKI being the most prevalent. Selective COX-2 inhibitors (celecoxib) have fewer gastrointestinal side effects, but their nephrotoxic profile is similar to nonselective NSAIDs.³² The onset of ATIN is usually after 6 to 18 months of therapy, and hypersensitivity syndrome is rare.³³

Renal histology shows interstitial inflammation and tubulitis, but the intensity is less pronounced compared with other DI-ATIN forms, and interstitial eosinophils may be absent.³³ Concomitant glomerular lesions can be observed, with minimal change disease being more common than membranous nephropathy.³⁴ Lower degrees of inflammation seen on histology may be attributed to the antiinflammatory properties of these drugs and diversion of arachidonic acid metabolites into alternative pathways,

suppressing immune function. Granulomatous lesions and TBM immune deposits have been reported with NSAIDs but are rare findings.³⁵ Despite severe AKI, complete recovery in NSAID-induced ATIN has been described in approximately 70% of pediatric studies.^{36,37}

ANTICANCER AGENTS

With the advancement of cancer therapies, the nephrotoxic potential of various agents has come to light. Among them, targeted agents have shown efficacy in treating multiple cancers by specifically targeting gene mutations in malignant tissue and inhibiting oncogenic signaling cascades associated with tumor growth. Immune checkpoint inhibitors (ICPIs) stand out due to their increasing use, treatment effectiveness, and immune-related side effects. According to a narrative review by the Cancer and Kidney International Network Workgroup on Immune Checkpoint Inhibitors, the incidence of AKI in ICPI-treated patients can be as high as 10% to 30%.³⁸ These inhibitors target ligands that bind inhibitory T-cell receptors, thereby activating host T cells and promoting infiltration of the tumor microenvironment and destruction of cancer cells.³⁹ However, this immune activation can lead to T-cell invasion of other organs, with the skin, gastrointestinal tract, and liver being the most commonly affected.²² Kidney involvement presents in the form of autoimmune acute tubulointerstitial nephritis. Among them, ipilimumab, a monoclonal antibody against CTLA-4, is approved for advanced malignant melanoma, while nivolumab and pembrolizumab, PD-1 inhibitors, are approved for advanced melanoma and squamous cell lung cancer.⁴⁰

The largest series to date, a multicenter study of 138 patients with immune checkpoint inhibitor-associated AKI, revealed that lower baseline eGFR, PPI use, and combination therapy were independent risk factors for AKI.²² The median time for AKI development was 14 weeks after initiating ICPIs (ranging from 6 to 37 weeks) and 2 weeks after the last dose. It is worth noting that nearly 70% of patients were taking concomitant medications known to cause interstitial nephritis including antimicrobials, NSAIDs, and PPIs.²²

Clinically, most patients presented with proteinuria >0.3 g/g on urine protein-to-creatinine ratio (71%) and pyuria on urine sediment ($\approx$ 50%). Peripheral eosinophilia, however, was not commonly observed (21%). Eighty-six percent of patients received corticosteroid treatment within 4 days of their serum creatinine doubling. Complete, partial, and no recovery of kidney function were observed in 40%, 45%, and 15% of patients, respectively.²²

Interestingly, rechallenge with ICPIs was attempted in 21% of cases, within an average of 1.8 months after the initial AKI episode. It is noteworthy that nearly 40% of these patients were receiving steroids at the time of rechallenge at doses ranging from 5 to 20 mg/day of prednisone. Unfortunately, 22% of patients experienced recurrent AKI within 1.4 months of rechallenge.²²

Before the ICPI era, the leading drugs associated with interstitial nephritis among cancer patients undergoing a kidney biopsy were ifosfamide (28%), bacillus Calmette-Guérin (12.5%), tyrosine kinase inhibitors (14%), and pemetrexed (9%).⁴¹ However, other drugs have been reported: BRAF inhibitors, bortezomib, platinum salts, lenalidomide, and many more. It is important to remember

that cancer drugs cause AKI through several mechanisms including acute tubular necrosis, glomerulonephritis, and thrombotic microangiopathy.⁴² For a more detailed discussion on cancer and the kidney, see Chapter 57.

IMMUNOSUPPRESSIVE THERAPY OF DRUG-INDUCED ATIN

The immunologic mechanism underlying ATIN and the presence of extensive inflammatory infiltrate on kidney biopsy suggest that immunosuppressive therapy may be a viable treatment option. However, the use of corticosteroids in DI-ATIN is a controversial topic because existing evidence is derived from retrospective studies rather than randomized controlled trials, making it less definitive. These studies, which relied on clinical judgment to decide on steroid use, were small and underpowered to detect potential benefits. Nevertheless, they suggest that initiating a trial of corticosteroids within the first week of discontinuing the causative drug provides the greatest benefit (Table 37.3).^{16,17,43-50} Late initiation of steroids has not shown improvement in kidney function, so in certain cases, drug withdrawal and steroid initiation are done simultaneously to prevent delays.

The initial approach involves discontinuing the offending agent. If kidney function fails to improve within 7 to 10 days of drug discontinuation, steroid therapy should be considered. Various corticosteroid regimens have been employed without significant differences in efficacy.^{43,49,50} While no standardized approach exists, prednisolone is commonly administered at a dose of 1 mg/kg/day (maximum 60 mg/day) for 2 to 3 weeks, followed by a tapering period lasting 5 to 6 weeks. The majority of kidney function recovery occurs within the first 4 weeks of treatment. In cases where benefits continue to be observed during the tapering period, longer treatment courses can be considered until kidney function reaches its baseline level. However, using steroids for more than 8 weeks in total or tapering off for more than 5 weeks does not provide additional recovery advantages.^{17,51} A multicenter, randomized, controlled clinical trial called PRAISE (PRednisolone Treatment Compared to Drug Withdrawal Alone in Acute Interstitial Nephritis) is under way to provide further insights.⁵²

The response to steroids varies on the basis of kidney biopsy findings. Patients with more than 50% interstitial fibrosis on biopsy exhibit poor response, whereas those with neutrophilic infiltrate tend to have a favorable response.^{48,49} Hence tissue diagnosis is crucial to not only initiate appropriate therapy but also evaluate the extent of scarring and potential for response. In cases where steroid therapy is not suitable, methotrexate, azathioprine, mycophenolate mofetil, or antitumor necrosis factor (anti-TNF) therapy can be used as steroid-sparing agents with promising outcomes.^{53,54}

SYSTEMIC DISEASES

AUTOIMMUNE DISORDERS

A variety of autoimmune disorders can lead to ATIN, more commonly in younger patients. The most common autoimmune diseases associated with interstitial nephritis include systemic lupus erythematosus (SLE), sarcoidosis, Sjögren syndrome, and tubulointerstitial nephritis and uveitis syndrome (TINU).⁵⁵ Autoimmune TIN has a

lymphocyte or plasma cell-rich infiltrate. Extraglomerular immune-complex deposition and immunofluorescence (IF) features are variable. Plasma cells are predominant in IgG4-related disease and Sjögren syndrome. Tubular basement membrane (TBM) immune-complex deposits are frequent in lupus, IgG4-related disease, and anti-LRP2 but are rare in Sjögren syndrome.

Lupus nephritis (LN) often presents with tubulointerstitial involvement, alone or in association with glomerular disease. While tubulointerstitial lesions have been reported in 50% to 70% of LN cases, current classifications focus primarily on glomerular lesions. Immune-complex deposits within the TBM drive the tubulointerstitial inflammatory lesion and are defined by positive stains for IgG, IgA, IgM, C1q, and C3 on immunofluorescence (“full house” pattern).^{55,56} While these deposits are typically found in the TBM, the interstitium or peritubular capillary walls can also be affected.⁵⁵ It is important to recognize that tubulointerstitial damage plays a significant role in predicting disease outcomes. In a study, 37% of patients with severe interstitial infiltrate progressed to kidney failure within 2 years.⁵⁶

Renal sarcoidosis manifests with various features including hypercalcemia, hypercalciuria, nephrocalcinosis, nephrolithiasis, and different glomerular lesions.⁵⁷ Granulomatous interstitial nephritis is the most common intrinsic kidney lesion seen in 80% of cases.⁵⁸ It is characterized by a diffuse inflammatory interstitial infiltrate that is composed of T lymphocytes, sometimes with noncaseating granulomas (Fig. 37.2). Steroids are an effective therapy, although complete response is not always achieved, and some patients experience partial remission or no improvement in kidney function, leading to CKD.^{57,58} Other immunosuppressive agents have been used in cases where steroids are not tolerated or are required long term.

Sjögren syndrome is a progressive autoimmune disorder primarily affecting the exocrine glands, such as the salivary and lacrimal glands. Kidney involvement is common, with estimates indicating it occurs in up to 40% of cases.³⁹ Interstitial nephritis is the most frequently observed kidney lesion reported in more than 70% of biopsy series.^{55,59,60} The primary feature of interstitial nephritis in Sjögren syndrome is lymphocytic infiltration, accompanied by tubular atrophy, tubulitis, and varying degrees of interstitial fibrosis. Tubular dysfunction manifests clinically as different tubulopathies, particularly distal RTA. The optimal treatment for interstitial nephritis in Sjögren syndrome remains uncertain. Immunosuppressive therapy including steroids, azathioprine, cyclophosphamide, rituximab, and hydroxychloroquine are commonly used.⁵⁵

Anti-LDL receptor-related protein 2 (LRP2) is an autoimmune disease manifested by IgG4 targeting brush border protein LRP2 (also known as “megalin”). The mean age of presentation is 73, and males are predominantly affected. LRP2 presents with AKI and subnephrotic levels of proteinuria, and patients progress to ESKF in 50% of cases. The IgG within the TBM deposits colocalizes only with LRP2 on confocal immunofluorescence, in contrast to other systemic diseases associated with IgG-containing TBM deposits, such as LN, IgG4-related kidney disease, and polyomavirus nephritis. Nonetheless, Anti-LRP2 nephropathy has been reported in patients with LN, minimal change disease, and low-grade B-cell lymphomas. Anti-B-cell agents such as rituximab or pulse

Table 37.3 Selected Studies of Corticosteroid Therapy in Acute Tubulointerstitial Nephritis

Author, Year	Design	N	Etiology	Arms	Steroid Regimen	Time to Steroid Treatment	Results
Clarkson et al., 2004 ¹⁷	Retrospective Single center 12-mo follow-up	42	92% Drugs 8% Idiopathic	16 Steroids 26 Conservative treatment	IV Methylprednisolone 500 mg/day 2–4 days Oral prednisolone 0.75 mg/kg/day tapered over 3–6 weeks	3.5 Weeks (mean time from symptom onset to biopsy)	No differences between groups in median serum creatinine at 1 year
Gonzalez et al., 2008 ⁴⁴	Retrospective Multicenter 19 mo follow-up	61	100% Drugs	85% Steroids	IV Methylprednisolone 250–500 mg/day 3–4 days Oral prednisolone 1 mg/kg/day tapered over 8–12 weeks	13 ± 10 Days in those with complete recovery vs. 34 ± 17 days in those with incomplete recovery ($P < 0.05$)	54% Complete recovery
Raza et al., 2012 ⁴⁵	Retrospective Single center 19-mo follow-up	49	67% Drugs Other: 4 TINU 2 Sarcoidosis 10 Idiopathic	75% Steroids	Oral prednisolone 1 mg/kg/day	2.9 days (time from hospital admission to biopsy)	Greater improvement in eGFR (3.4 vs. 2.1-fold; $P < 0.05$) Reduced need for dialysis (16 % vs. 42%; $P = 0.066$) 49% CR 39% PR
Muriithi et al., 2014 ¹⁶	Retrospective Single center 6-mo follow-up	133	71% Drugs 29% Other	86% Steroids	21% initially received IV steroids Starting dose of oral steroids was 40–60 mg/day, with a median treatment duration of 7.5 weeks	8 vs. 11 vs. 35 days in those with complete, partial and no recovery respectively; $P = 0.05$)	
Valluri et al., 2015 ⁴⁶	Retrospective Multicenter 12 mo follow-up	171	73% Drugs 10% Sarcoid/TINU 7% Autoimmune 8% Infection 2% Unclear	63% Steroids	Unknown	41% Early use (≤ 7 days) 48% Late (> 7 days) 11% unclear	No difference in CR (48% vs. 41%) Dialysis dependence was 10% in both groups
Ramachandran, et al., 2015 ⁴⁹	Prospective randomized Single center 12-week follow-up	29	100% Drugs	100% Steroids	Group 1: oral prednisolone 1 mg/kg/day for 3 weeks, followed by rapid tapering in the next 3 weeks. Group 2: IV methylprednisolone 30 mg/kg/day (max 1 g) for 3 days followed by oral prednisolone 1 mg/kg/day for 2 weeks and tapered in next 3 weeks	1 Week after withdrawal of causative drug in all cases	Oral and pulse steroid therapy equally effective in achieving remission

Continued

Table 37.3 Selected Studies of Corticosteroid Therapy in Acute Tubulointerstitial Nephritis — cont'd

Author, Year	Design	N	Etiology	Arms	Steroid Regimen	Time to Steroid Treatment	Results
Predecki et al., 2017 ⁴⁷	Retrospective Single center 39-mo follow-up	187	25% Drugs 48% Unknown 13% TB 9% Sarcoid 3% TINU 2% Sjögren	84% Steroids	Steroid therapy was determined by clinician choice: Prednisolone at doses of 40-60 mg/day Only 3 patients received methylprednisolone before oral prednisolone Group A: oral prednisolone 1 mg/kg/day for 2 weeks Group B: pulse methylprednisolone 30 mg/kg/day for 3 days (maximum 1 g), followed by oral prednisolone 1 mg/kg/day for 2 weeks, tapered over 2 weeks	NR	Steroid-treated patients had better eGFR at 2 years (43 vs. 24 mL/min) and less dialysis dependence (5.1% vs. 24.1%)
Chowdry et al., 2018 ⁵⁰	Prospective randomized 3-mo follow-up	31	100% Drugs	100% Steroids	Group A: oral prednisolone 1 mg/kg/day for 2 weeks Group B: pulse methylprednisolone 30 mg/kg/day for 3 days (maximum 1 g), followed by oral prednisolone 1 mg/kg/day for 2 weeks, tapered over 2 weeks	1 Week after withdrawal of causative drug	58% CR 42% PR There was no significant difference between the 2 groups.
Fernandez-Juarez et al., 2018 ⁴³	Retrospective Multicenter 6-mo follow-up	182	100% Drugs	100% Steroids	48% IV pulse steroids before oral therapy Initial dose of prednisone: 0.8 ± 0.2 mg/kg/day for 2 weeks	12 Days (mean interval between diagnosis and steroid initiation)	41% CR 46% PR
Surendra et al., 2019 ⁴⁸	Retrospective Single center 12-mo follow-up	83	100% Drugs	100% Steroids	500 mg Methylprednisolone for 3 days, followed by prednisolone of 0.5-1 mg/kg/day tapered over 4-6 weeks	5-7 days from drug withdrawal	47% CR 53% PR

CR, Complete recovery; DI-ATIN, drug-induced acute tubulointerstitial nephritis; eGFR, estimated glomerular filtration rate; NR, not reported; PR, partial recovery; sCr, serum creatinine.

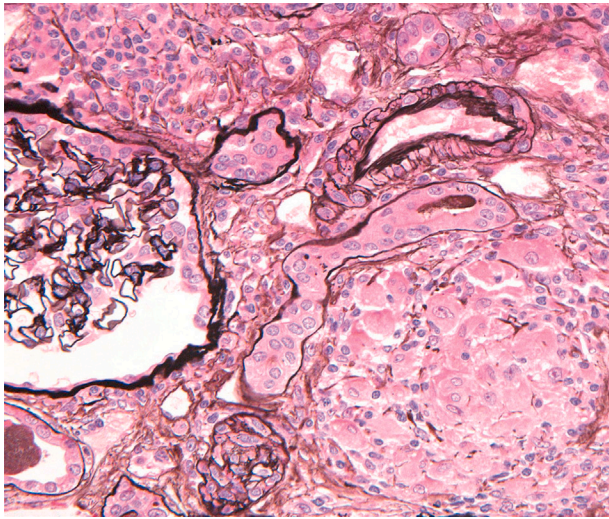


Fig. 37.2 Renal histology in sarcoidosis highlighting the inflammatory infiltrate and noncaseating granuloma. Jones silver stain. (From Perazella MA. Clinical approach to diagnosing acute and chronic tubulointerstitial disease. *Adv Chronic Kidney Dis.* 2017;24:57–63.)

corticosteroids with alkylating agents have shown variable success. However, immunologic remission has been associated with stabilization of kidney function after treatment with corticosteroids and cyclophosphamide.⁶¹

Tubulointerstitial nephritis with IgM-positive plasma cells (IgMPC-TIN) is an entity that was described in 2017.⁶² Its major clinical features include high serum IgM (s-IgM) levels, distal renal tubular acidosis (100%), Fanconi syndrome (92%), and antimitochondrial antibodies (82%).⁶² Treatment with glucocorticoids is effective, although relapse can occur.⁶³

Other immune-related disorders that primarily affect the renal microvasculature and glomeruli, such as immune-complex diseases (SLE or IgA nephropathy) or pauci-immune vasculitides (ANCA), may have additional ATIN findings. Granulomatosis with polyangiitis (GPA) and eosinophilic granulomatosis with polyangiitis (EGPA) are types of ANCA-associated vasculitides that can cause granulomatous TIN and periglomerular eosinophilic infiltrate. MPO-ANCA-positive cases often show more tubulointerstitial inflammation or scarring.⁵⁵

TUBULOINTERSTITIAL NEPHRITIS AND UVEITIS SYNDROME

Tubulointerstitial nephritis and uveitis (TINU) is a rare condition of unknown cause. The most comprehensive systematic review to date on 592 reported cases found that it predominantly affects females at a median age of 17 years.⁶⁴ Drugs and genetic factors, such as HLA associations (e.g., *DRB1*14*, *DQA1*01:04*, and *DQB1*05*), have been linked to susceptibility.^{55,64,65} Additionally, infectious triggers like tuberculosis, Epstein-Barr infection, herpes zoster, and *Campylobacter* have been implicated. TINU has also been observed in conjunction with relapsing polychondritis.⁵⁵

TINU presents as a clinical syndrome primarily involving the eyes and kidneys. Generalized, nonspecific symptoms such as weight loss, fever, anemia, and hypergammaglobulinemia are the initial manifestations. Kidney involvement can appear as isolated proximal tubulopathy or Fanconi syndrome, while uveitis is usually bilateral and anterior. Even

though it can occur before, during, or after the diagnosis of tubulointerstitial nephritis, uveitis often appears after the onset of renal symptoms (52% of cases).⁶⁴

Autoantibodies including ANCA, rheumatoid factor, anti-GBM antibody, and ANA are frequently observed in TINU patients. Renal histopathology commonly shows a predominance of CD3+ T lymphocytes, along with macrophages, plasma cells, and eosinophils. Granulomas are not common.⁵⁵

Steroids are the main treatment for both ocular and kidney manifestations. A 3- to 6-month course of prednisone is used, followed by a gradual taper to minimize the risk of relapse, which is relatively common. Other options to spare long-term steroid use include mycophenolic acid, methotrexate, cyclosporine, and cyclophosphamide. Adult age and the presence of posterior uveitis or panuveitis are associated with the development of CKD. However, ESKF is rare.⁶⁴

IgG4-RELATED KIDNEY DISEASE

IgG4-related kidney disease (IgG4-RKD) is a systemic fibroinflammatory disorder that commonly affects middle-aged men and involves various organ systems including salivary glands, pancreas, retroperitoneum, and kidneys. Kidney involvement occurs in approximately 18% of cases, often in conjunction with other affected organs.⁶⁶ Three different clinical phenotypes involving the kidney have been identified: obstructive uropathy from retroperitoneal fibrosis (the most common manifestation of IgG4-RKD), IgG4-related tubulointerstitial nephritis, and IgG4-related glomerular lesions, particularly membranous nephropathy.

IgG4-related tubulointerstitial disease is rare compared with retroperitoneal fibrosis. It presents with acute, subacute, or chronic kidney dysfunction and mostly bland urinalysis with a median proteinuria of 0.6 g/g on urine protein-to-creatinine ratio.⁶⁷ Elevated serum levels of IgG4 subclass antibodies and polyclonal hypergammaglobulinemia are seen in 94% and 90% of IgG4-RKD, respectively.⁶⁷ In addition, hypocomplementemia (45%), a positive ANA (36%), and ANCA (22%) have been reported.⁶⁷ It is important to note that elevated serum IgG4 levels are not specific to the disease because they can also be elevated in allergic, inflammatory, and neoplastic disorders. Imaging techniques such as contrast computed tomography (CT) or magnetic resonance imaging are abnormal in 61% of the cases, with pseudotumor (27%), low-density lesions primarily in the renal cortex (25%), and hypertrophy (19%) being among the most common findings.⁶⁷ Pathologic tissue in IgG4-RKD is characterized by a plasma cell-rich tubulointerstitial nephritis with a predominance of IgG4-positive plasma cells. Interstitial fibrosis with a “storiform” pattern that is typified by a cartwheel appearance of arranged fibroblasts and inflammatory cells known as “bird’s eye” is characteristic of this disease. Immunohistochemistry shows >10 IgG4-positive plasma cells per high-power field (40× lens), which translates to an increased IgG4/IgG plasma cell ratio (>40%).⁶⁸ Eosinophil infiltration is also frequent. TBM immune-complex deposits are found in 80% of cases and have nonspecific IF findings with positive IgG, kappa, and lambda, ±C3.⁶⁸ Other reported kidney lesions include granulomatous ATIN, masslike kidney lesions (Fig. 37.3).

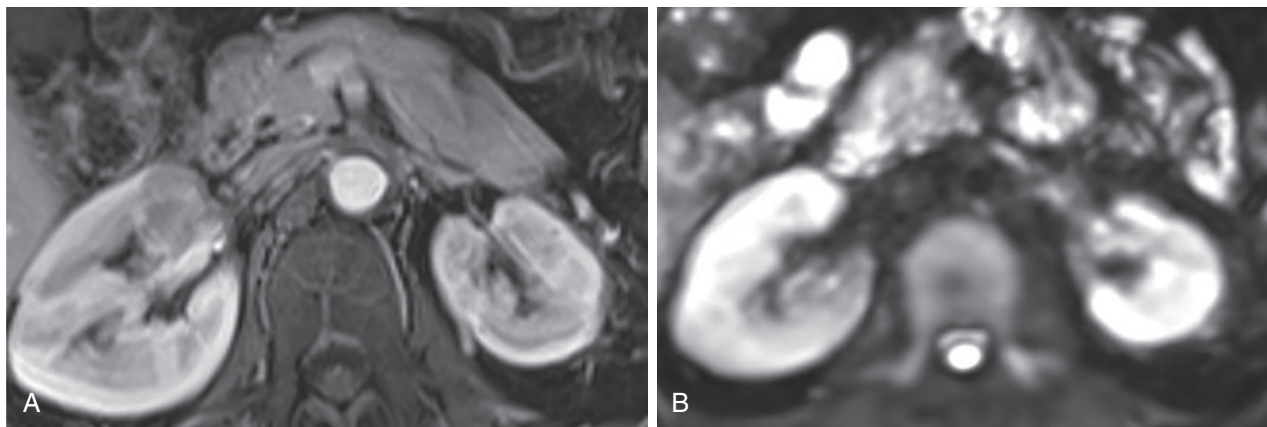


Fig 37.3 IgG4-related kidney disease emulating malignancy. (A) Magnetic resonance imaging (MRI) of the kidney with contrast showing multiple well-circumscribed right renal masses, the largest measuring $2.3 \times 1.9 \times 2.5$ cm, with imaging characteristics suggestive of a malignancy, such as papillary renal cell carcinoma or lymphoma. (B) MRI of the kidney shows resolution of the right renal-enhancing lesions after treatment with steroids. (Images courtesy Dr. Mohamed G. Atta.)

Treatment typically involves long-term glucocorticoid therapy, which generally yields favorable outcomes. However, a multidrug approach with agents like azathioprine, mycophenolic acid, or rituximab may be necessary.^{66,67,69} Prognosis varies, and even individuals with significant fibrosis may respond well to immunosuppressive therapies, although relapses are common. In a recent retrospective study of 101 IgG4-RKD cases, IgG4 concentrations (≥ 5 g/L) were associated with more severe kidney disease. Complement consumption and the number of organs involved were associated with higher relapse rates, while rituximab as a first-line therapy has shown lower relapse rates.⁶⁷ Recurrence after kidney transplantation has been reported.⁷⁰

INFECTIONS

Infection-related ATIN accounts for 5% to 10% of the cases, although its incidence seems to be higher in less developed countries.¹⁹ Bacteria invade the interstitium of the kidney through ascending infection or hematogenous spread, which is followed by an inflammatory response resulting in acute or chronic TIN. Bacterial pyelonephritis should be suspected when the interstitial infiltrate consists predominantly of neutrophils. Common bacteria associated with TIN include *Staphylococcus aureus* and *epidermidis*, *Salmonella typhi* and *paratyphi*, *Legionella* spp., *Yersinia enterocolitica*, *Brucella* spp., *Campylobacter jejuni*, and *Corynebacterium diphtheriae*.⁷¹ Additionally, granulomatous forms of interstitial nephritis have been reported with *Mycobacterium tuberculosis* and *leprae*.⁷¹

Hantaviruses (HVs), transmitted by infected rodents and bats, are the primary cause of viral-induced ATIN in immunocompetent individuals.^{72,73} In Eurasia, HV causes hemorrhagic fever with renal syndrome (HFRS), while in America, it leads to cardiopulmonary syndrome. The prevalent *Puumala hantavirus* serotype in Central and Northern Europe causes a milder form of HFRS known as “nephropathia epidemica,” characterized by flulike symptoms, AKI, and thrombocytopenia. Initially, patients present with oliguria and peak serum creatinine levels on day 7 of the disease.^{72,73} This is followed by polyuria and spontaneous recovery within 3 weeks. Kidney dysfunction

can present with hematuria, proteinuria (sometimes in the nephrotic range), and variable stages of AKI.^{72,73} The most specific pathologic finding of Hantavirus nephropathy is the presence of large areas of interstitial hemorrhage in the medulla.⁷⁴ However, this finding may not always be identified due to limitations in biopsy sampling. Interstitial nephritis is commonly observed in cortical samples, and in some cases, interstitial hemorrhage may extend to the cortex.⁷² The diagnosis of nephropathia epidemica is confirmed using routine enzyme-linked immunosorbent assay tests that detect IgG and IgM antibodies against HV. Currently, there is no specific antiviral treatment available for nephropathia epidemica, and management is symptomatic.^{72,73}

In HIV-positive individuals with AKI, 11% of cases are diagnosed with ATIN.⁷⁵ Medications are the most common cause, with NSAIDs, sulfonamides, PPIs, and antimicrobials being frequent culprits. Certain antiretroviral drugs like protease inhibitors (e.g., atazanavir and indinavir), alone or in combination with tenofovir alafenamide, have also been reported to cause ATIN.⁷⁵ Additionally, other rare systemic syndromes that are caused by HIV-induced immune dysregulation may present with interstitial nephritis. For example, immune reconstitution inflammatory syndrome (IRIS) with ATIN typically occurs when antiretroviral treatment is initiated in patients who are untreated or recently started on treatment for an opportunistic infection such as tuberculosis.⁷³ On the other hand, diffuse infiltrative lymphocytosis syndrome (DILS) presents as a Sjögren-like syndrome with parotid enlargement and distal renal tubular acidosis or ATIN and is typically seen in untreated HIV-positive individuals.⁷³ In the case of IRIS, kidney biopsy reveals acute granulomatous interstitial nephritis characterized by dense infiltrates of mononuclear cells, primarily composed of macrophages, CD4 T lymphocytes, and noncaseating granulomas. Treatment typically involves a short course of prednisone, antimicrobial therapy, and continuation of antiretroviral therapy, which can effectively reverse kidney injury. In contrast, in patients with DILS and ATIN, kidney biopsy is characterized by a lymphocyte-predominant infiltrate with CD3⁺CD8⁺ T lymphocytes (compared with CD4⁺ T lymphocyte infiltrate in classic Sjögren syndrome with tubulointerstitial nephritis), plasma

cells, and monocytes, without the presence of granulomas. Tubulitis and varying degrees of interstitial fibrosis and tubular epithelial atrophy, particularly affecting distal tubules, are commonly observed. While corticosteroids can effectively manage the manifestations of DILS, incomplete recovery or relapse may occur even with the use of antiretroviral therapy. Other forms of interstitial nephritis associated with HIV include plasmocytic interstitial nephritis showing CD138⁺ plasma cell infiltrate on kidney biopsy and infectious causes related to immunocompromised states.^{73,75}

Coronavirus disease 2019 (COVID-19) infection is frequently associated with kidney involvement, with acute tubular injury being the dominant lesion. Interstitial nephritis has also been reported in connection with both direct viral infection and mRNA vaccination.^{76,77}

Other viral infections causing ATIN include cytomegalovirus, adenovirus, BK virus, and Epstein-Barr virus, but these mainly affect transplant recipients. Additionally, ATIN has been associated with several other infectious agents including *Trypanosoma cruzi*, *Plasmodium malariae*, *Toxoplasma gondii*, *Leptospira*, *Leishmania*, *Histoplasma*, *Candida* species, *Mycoplasma*, *Rickettsia*, and *Babesia*. Histoplasmosis may present with noncaseating granulomatous interstitial nephritis.

In conclusion, various infections can lead to ATIN with different histologic patterns and clinical presentations. Proper diagnosis and management depend on identifying the underlying cause.⁷¹ For more discussion on infections and the kidney, see [Chapters 34 and 60](#).

ACUTE TUBULOINTERSTITIAL NEPHRITIS IN KIDNEY TRANSPLANTS

Viral infections including cytomegalovirus (CMV), BK virus, and adenovirus are the primary causes of TIN in kidney transplant recipients. Prophylactic measures have successfully reduced CMV infections after transplantation. BK virus-associated nephropathy (BKVAN) affects 1% to 10% of recipients, peaking within 2 to 6 months post-transplantation. Adenovirus primarily affects the urinary tract within the first 3 months in 5% of recipients. The risk of CMV infection is highest when the donor is CMV seropositive and the recipient is seronegative. Besides infections, acute rejection can cause TIN, commonly presenting within the first-year post-transplant. Later episodes are usually due to nonadherence or deliberate reduction of immunosuppression. Cellular rejection is characterized by mononuclear cell infiltration, occasionally with eosinophils. Neutrophils are not common in cellular rejection and suggest infection or antibody-mediated rejection. T-cell-mediated rejection (TCMR) diagnosis is based on the Banff criteria.⁷⁸

Treatment of viral infections in kidney transplant recipients involves reducing immunosuppression and antiviral therapy. Antimetabolites are discontinued and may be resumed later at lower doses for high-risk rejection patients. For patients with CMV, serial viral monitoring and valganciclovir prophylaxis are recommended. Those with BK virus nephropathy require discontinuation of antimetabolites indefinitely, and reduction of calcineurin inhibitors is advised if viremia persists. Intravenous immunoglobulin can be considered. In cases of adenovirus infection, treatment includes immunosuppression reduction, cidofovir, and intravenous immunoglobulin. Finally, TCMR treatment varies on the basis of Banff criteria, involving increased

immunosuppression and rabbit antithymocyte globulin in certain cases with minimal fibrosis and atrophy.⁷⁸ For a more in-depth discussion, see [Chapter 69](#).

IMAGING STUDIES

Imaging studies, such as renal ultrasound and CT of the abdomen, are not specific for ATIN but can be useful in identifying other causes of AKI like obstruction. A renal ultrasound is preferred in the initial workup due to its noninvasive nature and cost-effectiveness. A renal ultrasound and CT of the abdomen can reveal bilateral kidney enlargement and diffuse cortical hyperechogenicity attributed to cellular infiltration and edema.

In cases where a kidney biopsy carries high risk of bleeding, alternative imaging techniques like positron emission tomography (PET)-CT and gallium-67 scans can aid in detecting kidney inflammation. Gallium-67 has been used for ATIN diagnosis, but its accuracy is limited because it cannot distinguish between ATIN and other inflammatory conditions like glomerulonephritis. On the other hand, PET-CT scans are more specific because fluorodeoxyglucose accumulates in metabolically active inflammatory cells. However, the use of PET-CT in ATIN diagnosis has not been widely adopted.^{79,80}

NOVEL SERUM AND URINE BIOMARKERS

Other noninvasive tests for diagnosis of ATIN have been investigated. Given that ATIN is often a hypersensitivity reaction to medications, certain studies have examined cytokines released by CD4⁺ T helper cells. In one study involving 265 patients who underwent kidney biopsies for acute kidney disease evaluation, urine levels of interleukin (IL)-9 and tumor necrosis factor (TNF)- α were found to be independently associated with ATIN. These findings suggest that these cytokines could serve as potential diagnostic markers for ATIN.⁸¹ Additionally, another study identified and validated CXCL9, an IFN- γ -induced chemokine involved in lymphocyte chemotaxis, as a diagnostic biomarker for ATIN.⁸²

CHRONIC TUBULOINTERSTITIAL NEPHRITIS

CTIN is characterized by progressive, often indolent, loss of kidney function along with the histologic features of tubular atrophy, interstitial mononuclear cell infiltrates, and interstitial fibrosis. CTIN has numerous etiologies that we will review in detail ([Table 37.4](#)). Toxic and genetic factors are the most prominent ones.

ARISTOLOCHIC ACID-INDUCED INTERSTITIAL NEPHROPATHY (BALKAN AND CHINESE HERB NEPHROPATHY)

Aristolochic acid nephropathy (AAN) is a condition caused by compounds derived from plants belonging to the *Aristolochia* species, particularly *A. clematitis* and *A. fangchi*, which are known to be pathogenic.⁸³⁻⁸⁵ Balkan endemic nephropathy (BEN) was described in the 1920s as a tubulointerstitial disease affecting families living in the

Table 37.4 Causes of Chronic Tubulointerstitial Nephritis

Disease	Specific Causative Agents/Gene Mutations
Genetics	
Autosomal dominant tubulointerstitial kidney diseases (ADTKD)	ADTKD-UMOD: Uromodulin (<i>UMOD</i>) gene mutations in chromosome 16p12.3 ADTKD-MUC1: Mucin-1 (<i>MUC1</i>) mutations in chromosome 1q22 ADTKD-REN: Renin (<i>REN</i>) mutations in chromosome 1q32.1 ADTKD-HNF1 β : Hepatocyte nuclear factor 1- β (<i>HNF1β</i>) mutations in chromosome: 17q12 ADTKD-SEC61A1: <i>SEC61A1</i> mutations in chromosome 3q21.3
Cystinosis	<i>CTNS</i> gene mutations in chromosome 17p13
Karyomegalic interstitial nephritis	Fanconi anemia-associated nuclease 1 (<i>FAN1</i>) gene mutations
Nephronophthisis (NPHP)	Autosomal recessive mutations in the (<i>NPHP1-20</i> genes) that encode nephrocystin protein <i>NPHP1</i> is the most frequent
Others	<i>DNAJB11</i> , <i>PAX2</i> mutations
Toxic Factors and Potential Sources of Exposure	
Endemic nephritis	Itai-itai disease (Japan): Linked to cadmium exposure in water, crops, mining, pollution Balkan endemic nephropathy or Chinese herb nephropathy: Caused by aristolochic acid contaminating wheat CKDu Mesoamerican nephropathy Sri Lanka CKDu India CKDu
Heavy metal toxicity	Arsenic: Groundwater, manufacturing (plywood, others) Cadmium: Mining; environmental runoff Chromium: Mining, electroplating Lead: Mining, refining, paint; battery manufacture; moonshine Mercury: Manufacturing (batteries, light bulbs); cosmetics; traditional medicines Uranium: Mining; contaminated groundwater; warfare
Lithium	Treatment of neuropsychiatric conditions
Metabolic	Hypokalemia, oxalate, uric acid
Radiation exposure	In cancer treatments, occupational, nuclear plant accidents, warfare
Others	
Plants, mushrooms, and herbal supplements	<i>Cortinarius orellanus</i> and <i>Omphalotus orarius</i> , <i>Amanita smithiana</i> mushrooms, Aristolochic acid-containing plants, and medications
Vesicoureteral reflux	
Lymphoproliferative disorders, plasma cell dyscrasias	

CKDu, Chronic kidney disease of unknown etiology.

regions along the Danube River including Serbia, Bosnia, Herzegovina, Croatia, Romania, and Bulgaria. It was later discovered that BEN was associated with the ingestion of bread contaminated with seeds from *A. clematitis*, leading to a prolonged but low-grade exposure to aristolochic acid. Similarly, “Chinese herb nephropathy” emerged as a cause of progressive renal failure in the early 1990s, with *A. fangchi* (birthwort and Fang chi) used in weight loss preparations being identified as the culprit.⁸³⁻⁸⁵

Patients with AAN usually present with nonspecific symptoms and elevated serum creatinine levels. Anemia disproportionate to the degree of kidney dysfunction (due to destruction of peritubular erythropoietin-producing cells) and renal Fanconi syndrome are observed.⁸³⁻⁸⁵ Chinese herb nephropathy tends to progress more rapidly, while BEN has a slower progression, taking more than 20 years to develop ESKF.^{83,84} Histopathologic features for AAN

include hypocellular interstitial fibrosis decreasing from the outer to inner cortical regions and presence of DNA adducts in the renal tissue or urinary tract that can be identified in specialized laboratories.⁸⁵

Due to its strong association with nephropathy and uroepithelial malignancies, most health agencies worldwide have implemented bans on supplements containing aristolochic acid. However, the lack of regulatory oversight over supplements means that many *Aristolochia* extracts are still being used globally.^{83,84} Prompt diagnosis and management of AAN are crucial, including avoiding further exposure to aristolochic acid and considering glucocorticoid therapy. Patients with a history of AA exposure should undergo regular cancer surveillance: yearly CT scans, cystoscopy with ureteroscopy, and biannual urine cytology. In cases where kidney replacement therapy is initiated or before transplantation, bilateral nephroureterectomy

should be strongly considered.^{83,84} For further discussion on AAN, see [Chapter 78](#).

CHRONIC KIDNEY DISEASE OF UNKNOWN ETIOLOGY

Chronic kidney disease of unknown etiology (CKDu) is a condition that primarily affects young to middle-aged men working in agricultural fields. It was first reported in 2002, when a cohort of patients with advanced CKD without a clear etiology was observed in El Salvador.⁸⁶ Since then, a similar form of progressive CKD of unclear etiology has been reported in the regions of Meso-America (Nicaragua, Costa Rica, and Guatemala); Sri Lanka; India; Egypt; and Tunisia.^{66,83,87}

This particular form of CKD is characterized by 1. limited occurrence in the coastal Pacific area of Central America; 2. no association with known CKD causes such as diabetes mellitus; 3. a predominance of younger age and male gender; and 4. prevalence predominantly among low-income laborers working under extreme environmental conditions, specifically those involved in agricultural activities like cotton and sugarcane harvesting.⁸⁸ The disease typically presents as asymptomatic, progressive CKD, low-grade (<1 g/day) proteinuria, bland urine sediment, and occasionally mild hyperuricemia and hypokalemia.⁸⁸ Kidney biopsies often reveal tubulointerstitial disease, with some cases showing glomerular ischemia and glomerulosclerosis.⁸⁹

The exact etiology of CKDu remains unknown, although factors such as heat stress, pesticide exposure, consumption of water contaminated with agrochemicals and heavy metals, excessive use of NSAIDs, and genetic predisposition have been suggested.^{83,87} Heat stress with recurrent dehydration/volume depletion and potentially rhabdomyolysis episodes due to strenuous labor are believed to play crucial roles in the development of this condition.⁹⁰ In addition, in Sri Lankan and Indian patients, chronic exposure to cadmium and polymorphisms in *CYP1A1* (cytochrome P450, family 1, subfamily A), the *KCNA10* (voltage-gated K channel) gene, and other genes have been reported in CKDu cases.^{83,91-94} Progression to ESKF varies, but due to limited access to kidney replacement therapy in affected regions, mortality rates are high.⁸⁸

Currently, the best preventive measure for this condition is to ensure adequate hydration and minimize exposure to extreme heat.⁹⁰ Ongoing field studies are being conducted to further investigate the pathogenesis of this devastating condition. For more detailed discussion on CKDu, see [Chapter 83](#).

GENETIC CAUSES OF CHRONIC TUBULOINTERSTITIAL NEPHRITIS

The main genetic disorders associated with CTIN are reviewed here. For a more detailed discussion on Genetics of Kidney Diseases, please see [Chapter 44](#).

AUTOSOMAL DOMINANT TUBULOINTERSTITIAL KIDNEY DISEASES

Autosomal dominant tubulointerstitial kidney disease (ADTKD) is one of the most common genetic disorders of kidney disease accounting for approximately 5% of monogenic disorders causing progressive CKD. Many

different gene mutations linked to the disease account for the majority of suspected ADTKD cases, including *UMOD* gene encoding for uromodulin protein (Tamm-Horsfall protein), *REN* gene encoding for renin, *HNF1 β* gene encoding for hepatocyte nuclear factor 1 β , *MUC1* gene encoding for mucin-1 protein, and *SEC61A1* gene encoding for the pore-forming Sec61 α subunit of the Sec61 complex, which governs endoplasmic reticulum protein transport and passive calcium leakage.⁹⁵ *UMOD* and *MUC1* gene mutations are perhaps the most prevalent forms associated with ADTKD.^{96,97}

Clinical characteristics vary among the various forms of ADTKD. However, the most common presentation involves a family history of kidney disease given its autosomal dominant inheritance, progressive CKD, low-level proteinuria, and bland urinalysis.⁹⁸ Early-onset hyperuricemia or gout before the diagnosis of kidney disease is seen with *UMOD*, *REN*, and *HNF1B* mutations.⁹⁷ Specific features include hypotension, hyperuricemia, hyperkalemia, acidemia, and anemia during childhood in ADTKD-REN; bilateral kidney cysts, hypomagnesemia, pancreatic atrophy, and maturity-onset diabetes mellitus of the young type 5 (MODY5) in ADTKD-HNF1B; and congenital anemia, intrauterine growth retardation, preaxial polydactyly, and variable degrees of velopharyngeal insufficiency in ADTKD-SEC61A1, which is a rare mutation.⁹⁷ Kidney cysts are described in more advanced cases, but their prevalence is similar to other noncystic renal diseases.

Histology is characterized by interstitial fibrosis with tubular atrophy and thickened tubular basement membranes. Immunohistochemistry can identify the intracellular accumulation of the mutant protein. ADTKD-REN presents with an enlarged juxtaglomerular apparatus. To date, only ADTKD-UMOD can be diagnosed with a kidney biopsy. On electron microscopy, ADTKD-UMOD shows intracellular electron-dense granular material in the endoplasmic reticulum, representing accumulation of the mutated Tamm-Horsfall protein in the thick ascending limb and distal tubular cells.⁹⁷

Genetic testing, such as targeted gene panels or whole-exome sequencing, can detect most pathogenic variants associated with ADTKD including *UMOD* (in chromosome 16p12.3), *REN* (in chromosome 1q32.1), *HNF1B* (in chromosome 17q12), and *SEC61A1*. However, the most common pathogenic *MUC1* variant, a cytosine duplication in the variable number of tandem repeats region of the gene resulting in a frameshifted protein (*MUC1-fs*), cannot be detected through these methods.⁹⁷ Therefore if the ADTKD gene panel results are negative, targeted genetic testing should be performed to identify the *MUC1* variant. This specific *MUC1* genetic testing is available at the Broad Institute in Cambridge, Massachusetts and free worldwide via an international registry that is federated and whose U.S. principal investigator is Anthony J. Bleyer, M.D.⁹⁷

ADTKD patients with hyperuricemia or gout require allopurinol or febuxostat. Caution should be exercised when using diuretics because they may exacerbate hyperuricemia and volume depletion. Fludrocortisone is used to treat volume depletion and hyperkalemia in ADTKD-REN, while magnesium repletion is necessary for ADTKD-HNF1B. ADTKD-REN and ADTKD-SEC61A1 cases have higher rates of anemia and can benefit from erythropoietin therapy.⁹⁷

ESKF develops in ADTKD patients around 30 to 50 years of age, although the timing varies depending on the mutation.⁹⁷ The disease does not recur after kidney transplantation, which makes it an effective treatment option, but family members should be screened for mutations before donating an organ.

NEPHRONOPHTHISIS

Nephronophthisis (NPHP) is the most common monogenic cause of ESKD in individuals younger than the age of 30. It has an incidence ranging from 1 in 50,000 to 1 in a million. The most prevalent form is known as juvenile NPHP, which is linked to autosomal recessive mutations in more than 20 genes (*NPHP1-20*), which encode proteins involved in primary cilia, centrosome, and basal body functions. *NPHP1* homozygous deletion in chromosome 2q13 is the most frequent mutation.⁹⁹

The presentation of nephronophthisis can vary but typically includes urinary concentration defects with polyuria, polydipsia, and kidney cysts. Kidney disease has a gradual onset and is often diagnosed at advanced stages. Extrarenal involvement is common, affecting approximately 20% of *NPHP1* cases and 70% of those without *NPHP1*. Organs involved include the eyes, liver, heart, central nervous system, or bones, resulting in retinitis pigmentosa, hepatic fibrosis, situs inversus, cerebellar hypoplasia, and skeletal defects.⁹⁹

Diagnostic indicators of nephronophthisis include low urine osmolality and reduced hemoglobin and reticulocyte counts, which may be disproportionate to the degree of kidney dysfunction. Histologically, corticomedullary kidney cysts, CTIN, TBM disruption, and tubular atrophy are present. Genetic testing plays a crucial role in confirming the diagnosis and involves techniques such as polymerase chain reaction–based gel electrophoresis, next-generation sequencing gene panels for ciliopathy and tubulointerstitial kidney disease, or whole-exome sequencing. Copy number variation analysis can also detect whole gene deletions including the most common *NPHP1* variant.^{99,100}

Juvenile nephronophthisis can progress rapidly, leading to ESKF by the age of 13. Furthermore, it has been suggested that *NPHP1* mutations may contribute to cases of undiagnosed ESKF in adulthood.⁹⁹

CYSTINOSIS

Cystinosis is a rare genetic condition, affecting approximately 1 in every 100,000 to 200,000 live births.¹⁰¹ It is the most common cause of inherited renal Fanconi syndrome in children, which is almost always complete. Cystinosis can present in three clinical phenotypes. The most common is infantile nephropathic cystinosis, which accounts for 95% of cases and typically manifests as renal Fanconi syndrome, leading to progressive chronic kidney disease (CKD). The second type is juvenile or late-onset nephropathic cystinosis, characterized by milder symptoms of renal Fanconi syndrome or isolated proteinuria. Finally, the ocular or adult form primarily involves isolated corneal involvement. Cysteine accumulation in the cornea, thyroid, pancreas, gonads, and muscles leads to photophobia, diabetes, delayed puberty, male infertility, muscle atrophy, and decreased production of sweat, tears, and saliva among other symptoms.¹⁰¹

Corneal crystals start accumulating during the first year of life and are always present by 18 months of age on slit

lamp examination. Other clinical findings include growth retardation in infants born with normal height and weight. Kidney histology shows a lymphocytic and macrophage-predominant infiltrate in the cortex. Cystine crystals have a rectangular or rhomboid shape and are localized in the lysosomes on electron microscopy. Tubular epithelial cells have irregular shapes and can become atrophic at the tubular-glomerular junction, resulting in “atubular glomeruli,” also known as swan-neck deformity.¹⁰² Cystinosis is diagnosed on the basis of finding cystine corneal crystals on slit lamp examination, mutations in the *CTNS* gene located in chromosome 17p13, and high intracellular cystine levels in white blood cells. This is typically measured using either mixed leucocyte test or purified granulocyte fractions and can only be performed in specialized laboratories.^{101,103}

Treatment involves oral cysteamine, which enters the lysosomes and reacts with cystine to produce cysteine and a cysteine-cysteamine mixed disulfide that can exit the organelle using other lysosomal carriers. The use of cysteamine therapy can slow down CKD progression, but it cannot prevent ESKF. Also, it has minimal effect on the renal Fanconi syndrome, so electrolyte replacement is needed. In infancy, indomethacin can be used to reduce urinary volume losses. Because systemic therapy does not reach the cornea, cysteamine eye drops are used to prevent corneal crystal deposition.^{101,103}

If cystinosis patients do not receive cysteamine therapy, they tend to develop ESKF by the age of 10. With treatment, kidney survival improves for an additional 7 to 10 years.¹⁰⁴ Patients may require treatment for other complications of cystinosis, such as hypothyroidism and diabetes mellitus.^{101,103}

OTHER GENETIC DISORDERS

Heterozygous *DNAJB11* mutations can lead to slow progressive CKD, kidney fibrosis, and various kidney cysts, sometimes misdiagnosed as autosomal dominant polycystic kidney disease or ADTKD.¹⁰⁵ In contrast, homozygous *DNAJB11* mutations cause Ivermark II syndrome, characterized by severe oligohydramnios, interstitial fibrosis, and kidney cysts. Tubulointerstitial kidney disease can also result from *PAX2* and mitochondrial DNA mutations, where ocular findings assist in distinguishing *PAX2*-related cases from ADTKD.¹⁰⁶ Moreover, mutations in the Fanconi anemia–associated nuclease 1 (*FAN1*) gene are linked to karyomegalic interstitial nephritis, a rare autosomal recessive disease. While sharing similar histology with NPHP, karyomegalic interstitial nephritis is characterized by the presence of karyomegaly in the kidneys, lungs, liver, and brain.¹⁰⁷

ANALGESIC NEPHROPATHY

Analgesic-induced CTIN, also known as “analgesic nephropathy,” is a condition associated with long-term ingestion of large quantities of combination analgesic medications. It is more common among women with chronic pain syndromes. Epidemiologic data vary among different countries and have been influenced by changes in analgesic usage patterns. Before the removal of phenacetin from analgesic mixtures between 1960 and 1980, analgesic nephropathy was a common cause of CKD in Scotland,

Belgium, and Australia, accounting for 10% to 20% of ESKF cases.¹⁰⁸

Combination analgesics containing aspirin or antipyrene with phenacetin, acetaminophen, salicylamide, caffeine, or codeine have been implicated in analgesic nephropathy, which is generally associated with prolonged and regular ingestion of at least 6 tablets daily for more than 3 years.¹⁰⁹ Although the specific causative agent remains unidentified, most data suggest that phenacetin is the culprit for analgesic nephropathy. Phenacetin has been removed from the market in most countries, resulting in decreased rates of this condition.¹¹⁰ Long-term aspirin or acetaminophen (paracetamol) monotherapy has not been directly associated with higher risks of CKD.^{111,112} While heavy NSAID use does not show a consistent association with CKD, chronic repeated intake of NSAIDs may increase the progression of CKD in patients with preexisting kidney disease.¹¹³

The medulla and papillary tip have the highest concentration of these drugs and their metabolites, resulting in cellular injury through lipid peroxidation and eventual capillary sclerosis.¹¹⁴ The use of aspirin can decrease glutathione levels in the kidney cortex, thereby augmenting the risk of acetaminophen toxicity. Additionally, aspirin and other NSAIDs inhibit the production of vasodilatory prostaglandins, potentially contributing to renal toxicity by reducing renal blood flow to the medulla.¹¹⁴

Patients present with generalized symptoms, nocturia, sterile pyuria, azotemia, and anemia. Imaging findings include small kidneys with bumpy renal contours and papillary calcifications on noncontrast CT scans, but they are not always present. There are no specific histologic features of this condition. Diagnosis is often made by excluding other causes of progressive kidney disease in CKD patients with a history of heavy analgesic use.

Discontinuation of analgesic use can slow disease progression, but recovery of kidney function is unlikely. Long-term use of these medications has been associated with urinary tract cancers in some patients.¹¹⁵

HEAVY METAL TOXICITY

Heavy metal toxicity through occupational or environmental exposure has been associated with CTIN (Table 37.4). Lead and cadmium have been described with more frequency, although nephrotoxicity from arsenic, mercury, and uranium has also been reported.^{116,117}

LEAD

Epidemiologic studies have strongly linked excessive lead exposure to CTIN, which can progress to kidney failure.¹¹⁸⁻¹²⁰ While occupational and environmental lead exposure has decreased, low-level exposure still occurs through sources like old water pipes, pottery, crystal, and lead-based paint in older homes.¹¹⁷ Notably, larger-scale lead exposures have been seen in contaminated water supplies, such as the situation in Flint, Michigan, in 2014.¹²¹

To accurately detect lead exposure, it is important to take a thorough history and measure total body lead burden through bone x-ray fluorescence studies and EDTA mobilization tests. Blood lead levels can be helpful but only in cases of recent exposure.¹²² Analysis of data from the Third National Health and Nutrition Examination Survey

(NHANES III) showed that even low-level lead exposure is associated with an increased risk of CKD among adults with hypertension in the United States, and in adolescents, the higher the blood lead levels, the lower the eGFR.^{123,124}

The pathogenesis of chronic lead nephropathy involves the reabsorption of filtered lead by the proximal tubule, resulting in Fanconi syndrome. Additionally, these patients present with hypertension and gout.^{117,125} The progression to ESKF is gradual and therefore chelation therapy with intravenous EDTA has been used with positive outcomes,^{122,126} though this still requires confirmation in larger studies.

CADMIUM

Cadmium can accumulate in the body through ingestion or inhalation. Prolonged low-level exposure, such as in zinc smelterworkers, or significant environmental contamination, as seen in Japan in the early 20th century in relationship to mining, can lead to cadmium nephropathy.^{116,127} Cadmium is bound to metallothionein, and proximal tubular cells take up the complexes leading to tubular dysfunction in the form of Fanconi syndrome and progressive, irreversible CKD.¹²⁸

The Cadmibel Study showed a direct correlation between urinary cadmium excretion and markers of renal tubular dysfunction. A 5-year follow-up study of individuals with the highest urinary cadmium excretion did not demonstrate evidence of progressive kidney damage or loss of function.^{116,129} Increased urinary calcium excretion seen in those exposed to cadmium has been associated with calcium phosphate kidney stones.¹³⁰

METABOLIC CAUSES

Hypokalemia, hyperuricemia, and hyperoxaluria have been associated with chronic interstitial nephritis through several mechanisms. In hypokalemic nephropathy, the alternative complement pathway is activated, while in cases of hyperuricemia and hyperoxaluria, inflammation occurs in the interstitium due to crystal formation.¹³¹

Hypokalemic chronic interstitial nephritis is more prevalent in women and is often linked to conditions like anorexia nervosa.^{131,132} Other related conditions include Bartter and Gitelman syndromes, as well as primary or secondary forms of mineralocorticoid excess. The development of chronic hypokalemia is thought to trigger ammonia production in the proximal tubule, resulting in the accumulation of ammonium in the medulla, complement activation, and fibrosis. Histopathologic findings include lymphocytic cellular infiltrate, vacuolar formation in tubular cells, and hypertrophy and hyperplasia of the juxtaglomerular apparatus.¹³¹ Treatment involves addressing the underlying condition and using potassium repletion or potassium-sparing diuretics. A 21-year follow-up study of patients with anorexia nervosa revealed that 5.2% developed ESKF after diagnosis.¹³²

Oxalate nephropathy can be classified as primary or secondary. Primary hyperoxaluria is a genetic disorder caused by enzymatic defects, with the most common form resulting from a deficiency in the vitamin B₆-dependent hepatic peroxisomal enzyme called alanine glyoxylate aminotransferase.¹³³ Secondary forms are caused by fat malabsorption (75%), followed by high dietary oxalate

intake, and less frequently decreased intestinal oxalate.¹³⁴ Oxalate is filtered by the glomeruli and secreted by the proximal tubules, leading to crystal formation and interstitial inflammation mediated by the nucleotide-binding domain leucine-rich repeat inflammasome (NALP3).¹³¹ Clinical presentation includes nephrolithiasis and progressive CKD. Treatment options involve supportive measures for kidney stones, reducing oxalate consumption, and considering new therapies like lumasiran and pyridoxine.^{133,135} The average age of onset for oxalate nephropathy is 9.5 years for primary hyperoxaluria and 56 years for secondary cases.^{133,134}

Uric acid nephropathy is associated with hyperuricemia and uric acid crystal formation, which activate inflammatory pathways such as the NALP3- inflammasome and nuclear factor- κ B (NF- κ B) signaling pathways.¹³¹ Kidney biopsy reveals tubulointerstitial fibrosis and crystal deposition, with a predominant granulomatous reaction observed in the medulla. Hyperuricemia and gouty arthritis are common clinical features, and medullary cysts have been reported in 30% to 70% of patients.¹³¹ The effectiveness of uric acid-lowering therapies in treating chronic urate tubulointerstitial nephropathy has not been proven.¹³⁶⁻¹³⁸

LITHIUM

Approximately 15% to 20% of patients who take lithium chronically may develop CKD during their treatment.¹³⁹ Prolonged exposure to therapeutic levels of lithium can double the decline in eGFR associated with aging.¹⁴⁰ The development of ESKD typically occurs after several decades of treatment. Alongside tubulointerstitial nephropathy (TIN), many cases of lithium-induced nephropathy show evidence of microcyst formation, which can be observed through histologic examination and kidney imaging studies (Fig. 37.4).¹¹⁷ Managing lithium-induced TIN involves discontinuing the use of lithium. However, this may not always be feasible since lithium is an effective therapy in controlling neuropsychiatric symptoms. In some cases, amiloride can be considered to reduce the risk of lithium-induced nephrotoxicity.¹⁴¹

RADIATION

Radiation nephritis can develop within 3 to 12 months after exposure to radiation, with chronic symptoms appearing approximately 18 months later. Hypertension, anemia, and renin-mediated hypertension are common.¹¹⁷

VESICoureTERAL REFLUX

Vesicoureteral reflux (VUR) is characterized by an abnormal uretero-vesical junction, leading to the backward flow of urine from the bladder to the kidneys. VUR has been implicated as a cause of CTIN. This condition is more prevalent in infants with a family history, occurring in approximately 20% of cases compared with 1% to 2% in the general population. Several genes including *PAX2*, *EYA1*, *GREM1*, *UPK3A*, *UPK2*, *HNF1B*, *ROBO2*, *SIX2*, *BMP4*, *SOX17*, and *TNXB* may contribute to VUR as a complication of congenital anomalies of the kidney and urinary tract (CAKUT).¹⁴²

VUR is often suspected in children presenting with recurrent urinary tract infections, along with additional

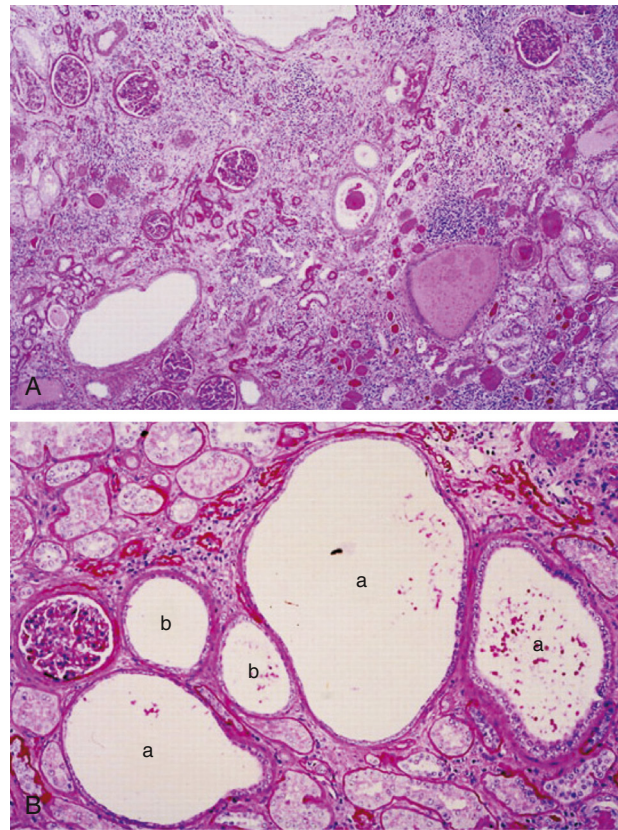


Fig. 37.4 (A) Severe lithium-associated chronic tubulointerstitial nephropathy, with the additional finding of focal tubular cysts arising in a background of severe interstitial fibrosis and tubular atrophy. (Periodic acid-Schiff [PAS] stain, $\times 40$.) (B) High-power view of tubular cysts lined by simple cuboidal epithelium (a). Adjacent tubules show tubular dilation (b). (PAS stain, $\times 100$.) (From Markowitz GS, Radhakrishnan J, Kambham N, et al. Lithium nephrotoxicity: a progressive combined glomerular and tubulointerstitial nephropathy. *J Am Soc Nephrol*. 2000;11:1439–1448.)

symptoms such as hypertension, polyuria, nocturia, constipation, and encopresis. Renal ultrasound can aid in the diagnosis by revealing hydronephrosis, upper urinary tract dilatation, or urinary tract malformations. While the gold standard for assessing VUR severity is a voiding cystourethrogram, this test can be challenging to perform in children.¹⁴²

CONCLUSIONS

In conclusion, patients with TIN come to clinical attention with loss of kidney function but without other specific symptoms or signs. To establish the diagnosis, the clinician must keep a high index of suspicion for this disease in all patients with acute or subacute loss of kidney function and gather careful history, as well as blood and urine testing. A kidney biopsy is required to establish the diagnosis. Identifying and stopping the offending agent is the first approach, and this can be followed by a course of steroids. Many novel tests are currently being evaluated and may be available for use in the near future.

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